[Date]

Editor-in-Chief

Health Care Management Science

Dear Editor,

I am submitting the manuscript "When Does Appointment-Scheduling Complexity Pay? Service-Time Uncertainty and the Value of Simple Planning Rules" for consideration as an Original Research article in Health Care Management Science, and I suggest the Health Care Operations Management department. The manuscript has not been published and is not under consideration elsewhere. [AUTHOR TO CONFIRM]

The paper addresses a decision that every outpatient clinic makes and that rarely gets a quantitative answer: how long the standard appointment interval should be, and how much more scheduling sophistication than that is worth buying. Many clinics still set the interval equal to the mean consultation time. I hold that mean fixed and vary only the shape of the service-time distribution (gamma, Weibull, lognormal, mixtures and Pareto-type tails). I then ask three separate questions: (A) what mean-only slots cost; (B) what a wrong assumption about the service-time family costs; and (C) what each additional degree of freedom in the interval vector is worth.

The answers are directly usable by clinic managers. (A) At a session of 30 patients, mean-based fixed slots carry 36.7%–71.3% relative cost regret against a computational six-block sample-average-approximation benchmark. (C) A single optimized uniform interval comes within 1.4% of that benchmark; ten interval blocks capture 91–97% of the remaining gap to a full 29-parameter design, and that residual value sits almost entirely in the first and last few positions. (B) Calibration matters more than complexity: under-designing the interval incurs on average about 3.7 times the excess cost of over-designing it, mainly when variability is ignored altogether, and a genuinely heavy-tailed truth is not resolved by moment fitting even with 1,000 historical observations. Constructed counterexamples (nonstationary service, known patient heterogeneity, dominant overtime cost) show where richer schedules do pay. The managerial message is therefore specific: measure the clinic's own consultation-time distribution, calibrate one interval, and invest in slot-by-slot design only where service is heterogeneous in a known, structured way.

I want to be precise about the novelty boundary. The near-optimality of constant job-allowance policies, and the fact that optimal interval vectors vary mainly at the start and end of a session, were established by Zhou, Ding, Huh and Wan (2021, Production and Operations Management) with finite-horizon bounds and numerical evidence. My paper builds on that result rather than rediscovering it. What it adds is a quantified map from distributional family, and from misspecification of that family, to the value of added

schedule complexity under an overtime-inclusive cost, together with the data required to act on it. The closest recent article in this journal, Khalilpourazari and Hashemi Doulabi (2026), develops a new stochastic program that solves sessions of up to 14 patients exactly under patient-and-time-dependent service, punctuality and no-show distributions. That is a method for heterogeneous patients. My paper offers no new algorithm. It shows when such machinery is and is not needed: for a homogeneous session the scalar interval suffices, while structured heterogeneity is one of the regimes in which complexity pays. The paper also connects to Shi et al. (2023) in this journal, who showed that oversimplified models of long-tailed durations are costly in surgical scheduling. I quantify that risk, and its asymmetry, for a single clinic session.

The evidence is computational and synthetic. I make no claim of observed patient-wait reductions, financial savings or external clinical validation, and the manuscript states the model's limits (single provider, homogeneous patients, punctual arrivals in the base case). All results, figures and tables are generated by publicly available code at https://github.com/bougtoir/tailaware-outpatient-scheduling (commit 20b691aa2b04eeb45db7b099cccff457b849a8c2). The code was reproduced from an anonymous fresh clone. No clinical data were used. Because the journal uses double-blind review, the repository address appears only on the title page and in this letter.

I used an AI software-engineering agent to help develop code, figures and drafts under my direction, as disclosed in the Methods section and the declarations. I checked the outputs and take responsibility for the content. [AUTHOR TO CONFIRM]

Thank you for considering this submission.

Sincerely,

Onishi Tatsuki

Data Science and AI Innovation Research Promotion Center, Shiga University, Hikone, Shiga, Japan [AUTHOR TO CONFIRM]

[e-mail: AUTHOR TO PROVIDE] ORCID 0000-0001-7261-9062